

PURPOSE

Characteristic crack width for a reinforced concrete slab under quasi-permanent loading.

1 Input Parameters

Symbol	Value	Unit	Description
b	1000	mm	Section width (per metre strip)
h	300	mm	Section depth
d	260	mm	Effective depth
c_{nom}	30	mm	Nominal concrete cover
ϕ	16	mm	Bar diameter
A_s	1005	mm^2	Tension reinforcement area (5 H16)
M_{qp}	45	kNm	Quasi-permanent bending moment
E_s	200000	MPa	Steel modulus of elasticity
E_{cm}	33000	MPa	Concrete secant modulus
$f_{ct, eff}$	2.9	MPa	Effective concrete tensile strength
k_t	0.4	-	Load duration factor (long term)
k_1	0.8	-	Bond factor (high-bond bars)
k_2	0.5	-	Strain distribution factor (bending)
k_3	3.4	-	Code coefficient
k_4	0.425	-	Code coefficient
w_{max}	0.3	mm	Allowable crack width (XC exposure)

2 Calculation

Modular ratio E_s/E_c cm

$$\alpha_e = \frac{E_s}{E_{cm}} = \frac{200000}{33000} = \mathbf{6.061}$$

Neutral axis depth (cracked section)

$$\begin{aligned} X &= \frac{A_s \cdot \alpha_e \cdot \left(\sqrt{\frac{2 \cdot b \cdot d}{A_s \cdot \alpha_e} + 1} - 1 \right)}{b} \\ &= \frac{1005 \cdot 6.061 \cdot \left(\sqrt{\frac{2 \cdot 1000 \cdot 260}{1005 \cdot 6.061} + 1} - 1 \right)}{1000} \\ &= \mathbf{50.52 \text{ mm}} \end{aligned}$$

Internal lever arm

$$\begin{aligned}
 z &= d - \frac{x}{3} \\
 &= 260 - \frac{50.52}{3} \\
 &= \mathbf{243.2 \text{ mm}}
 \end{aligned}$$

Steel stress under quasi-permanent loads

$$\begin{aligned}
 \sigma_s &= M_{qp} \cdot 1000000 \cdot \frac{1}{A_s \cdot z} \\
 &= 45 \cdot 1000000 \cdot \frac{1}{1005 \cdot 243.2} \\
 &= \mathbf{184.1 \text{ MPa}}
 \end{aligned}$$

Effective tension zone height

$$\begin{aligned}
 h_{c, eff} &= \min \left(\frac{h}{2}, \frac{h-x}{3}, 2.5 \cdot (-d + h) \right) \\
 &= \min \left(\frac{300}{2}, \frac{300-50.52}{3}, 2.5 \cdot (-260 + 300) \right) \\
 &= \mathbf{83.16 \text{ mm}}
 \end{aligned}$$

Effective reinforcement ratio

$$\begin{aligned}
 \rho_{p, eff} &= \frac{A_s}{b \cdot h_{c, eff}} \\
 &= \frac{1005}{1000 \cdot 83.16} \\
 &= \mathbf{0.01208}
 \end{aligned}$$

Maximum crack spacing

$$\begin{aligned}
 S_{r, max} &= c_{nom} \cdot k_3 + \frac{k_1 \cdot k_2 \cdot k_4 \cdot \phi}{\rho_{p, eff}} \\
 &= 3.4 \cdot 30 + \frac{0.425 \cdot 0.5 \cdot 0.8 \cdot 16}{0.01208} \\
 &= \mathbf{327.1 \text{ mm}}
 \end{aligned}$$

Mean strain difference (eps_sm - eps_cm)

$$\begin{aligned}
 \varepsilon_{diff} &= \max \left(\frac{0.6 \cdot \sigma_s}{E_s}, \frac{\frac{f_{ct, eff} \cdot k_t \cdot (\alpha_e \cdot \rho_{p, eff} + 1)}{\rho_{p, eff}} + \sigma_s}{E_s} \right) \\
 &= \max \left(\frac{0.6 \cdot 184.1}{200000}, \frac{184.1 - \frac{0.4 \cdot 2.9 \cdot (0.01208 \cdot 6.061 + 1)}{0.01208}}{200000} \right) \\
 &= \mathbf{0.0005524}
 \end{aligned}$$

Characteristic crack width

$$\begin{aligned}
 w_k &= \varepsilon_{diff} \cdot S_{r, max} \\
 &= 0.0005524 \cdot 327.1 \\
 &= \mathbf{0.1807 \text{ mm}}
 \end{aligned}$$

3 Verification

OK

$$w_k = 0.1807 \text{ mm} \leq w_{max} = 0.3 \text{ mm}$$